Number: P3309R3
Date: 2024-11-20
Audience: LWG
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atomic and atomic_ref

Introduction and motivation

This paper proposes marking most of `atomic<T>` methods and associated functions `constexpr` to allow usage of atomic code without changes in a constexpr and consteval code.

Proposed changes will allow implementing other types (`std::shared_ptr<T>`, persistent data structures with atomic pointers) and algorithms (thread safe data-processing, like scanning data with atomic counter) with just sprinkling `constexpr` to their specification.

Changes

- R2 → R3: Added location of library feature test macro, removed `atomic<shared_ptr>` and `atomic<weak_ptr>`.
- R1 → R2: Added clarification for behaviour of `wait` and `notify` functions.
- R0 → R1: Make `wait` and `notify` functions as requested by SG1. Wording changed accordingly. Updaded link to implementation on Compiler Explorer.

Previous polls

SG1: Forward P3309 to LEWG with the following notes:

- *Add constexpr to the wait and notify functions in the next revision of P3309*
- *`atomic<shared_ptr>` should be supported in constexpr whenever `shared_ptr` is supported in constexpr (whichever paper lands second should have this change)*
- *`is_lock_free()` should not be made constexpr*

SF	F	N	A	SA
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